Supplementary Material

Return-aware proximal policy optimization with pointer attention for multi-resource mountain-road unmanned aerial vehicle inspection planning

S1. Frozen protocol and reproducibility scope

This supplement reports secondary diagnostics, additional baseline views, failure-mode summaries, all learning curves, and the post-hoc operational composite. It does not change the confirmatory safe priority-weighted coverage outcome, map-level independent unit, or v3.2.14 protocol. The source result file contains 21,648 records, the final audit passed, and ppo_mlp is absent.

Table S1. Frozen audit facts.

Item	Frozen value	Claim boundary
Learning checkpoints	35 paper models	Seven variants × five seeds
Paper training episodes	105,000	Historical excluded work not used
Unseen procedural maps	24	Independent map-level test units
Geographic DSM regions	8	Simulation transfer only
Final route records	21,648	No manuscript rerun
Hierarchical bootstrap	10,000 replicates	Outer unit is map

S2. Performance profiles and solver diagnostics

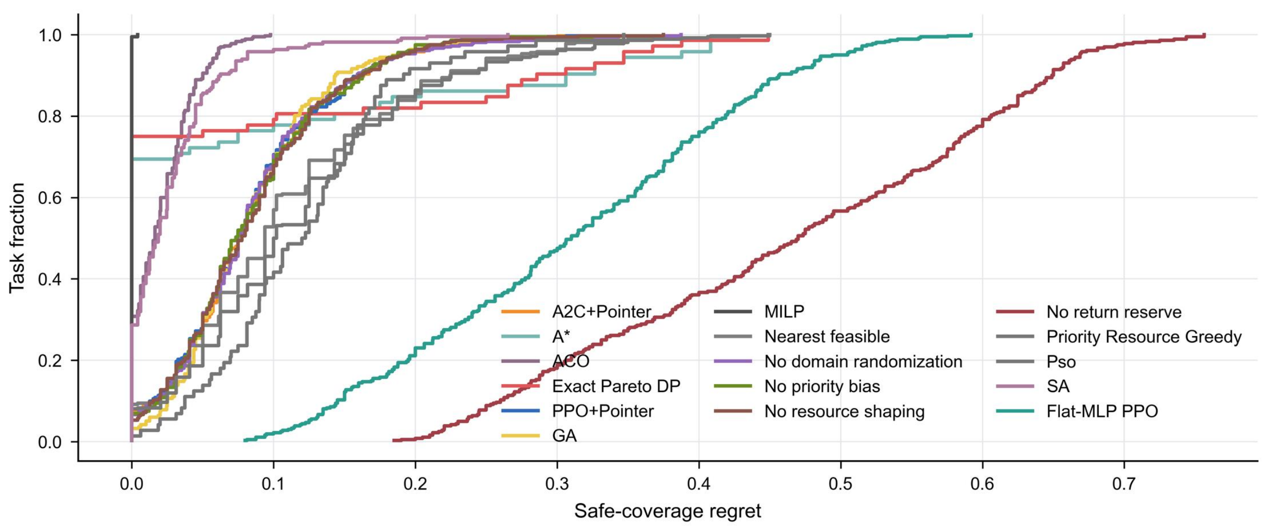

Fig. S1. Performance profiles based on regret from the safe priority-weighted coverage leader for each task.

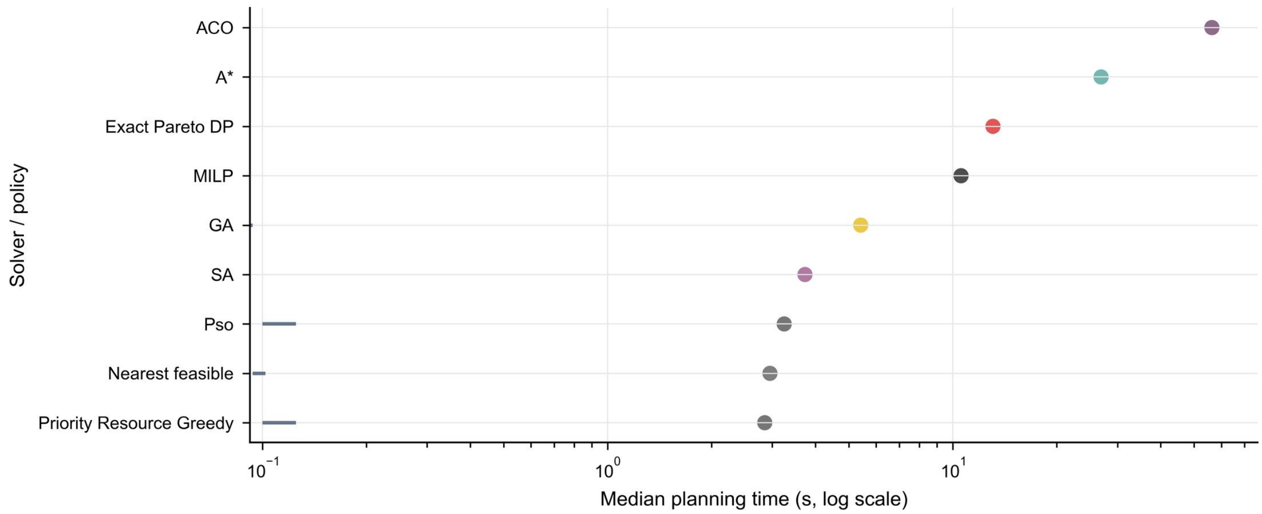

Fig. S2. Oracle regret and online computation. Horizontal intervals show the frozen regret range; planning time is logarithmic.

S3. Scenario-stratified and failure-mode results

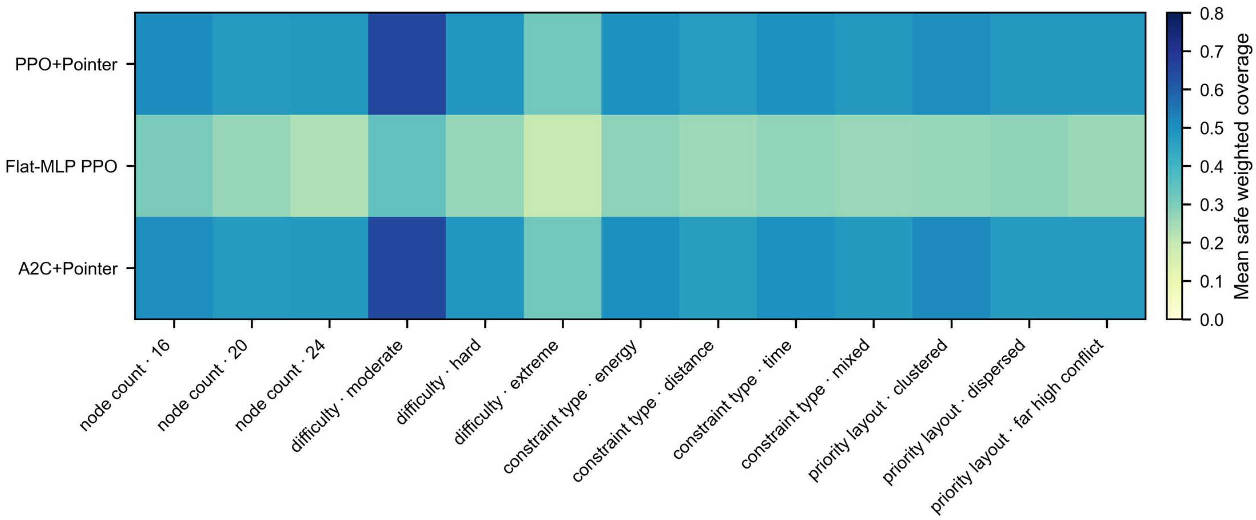

Fig. S3. Scenario-stratified safe priority-weighted coverage across node count, terrain, wind, and resource factors.

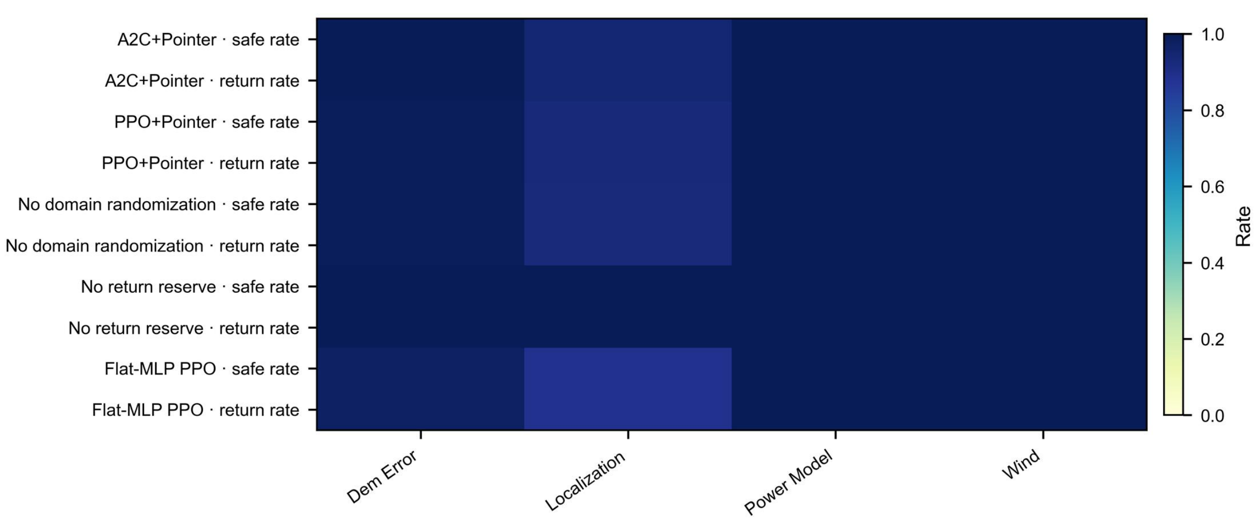

Fig. S4. Safety and return outcomes across robustness families and perturbation conditions.

S4. All-model training evidence

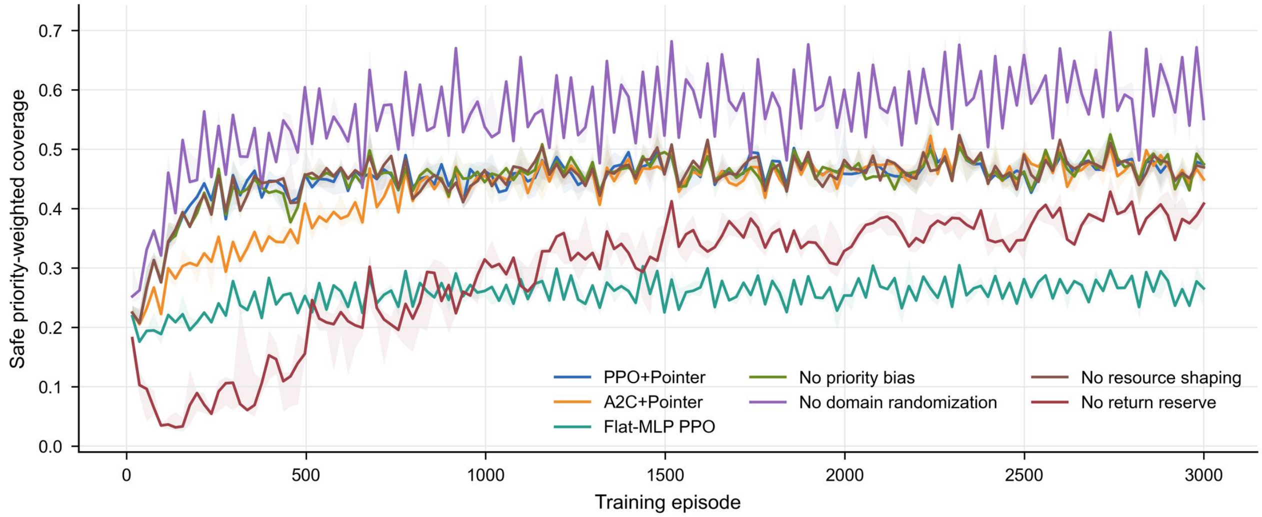

Fig. S5. Training trajectories for all seven learning variants under the frozen 3,000-episode protocol.

S5. Post-hoc operational summary and sensitivity

The seven-dimensional operational score combines D1–D7 after the confirmatory analyses. The arithmetic scores are 76.66 for PPO+Pointer, 74.55 for A2C+Pointer, and 54.47 for Flat-MLP PPO. A 10,000-replicate hierarchical bootstrap gives a mean PPO-minus-A2C difference of 1.99 points (95% interval 0.78–3.21; probability positive 1.00). This score is descriptive: its weights, operational floor, and normalisation do not replace safe priority-weighted coverage, and no individual D4, D6, or D7 contrast remained significant after Holm adjustment in its post-hoc family.

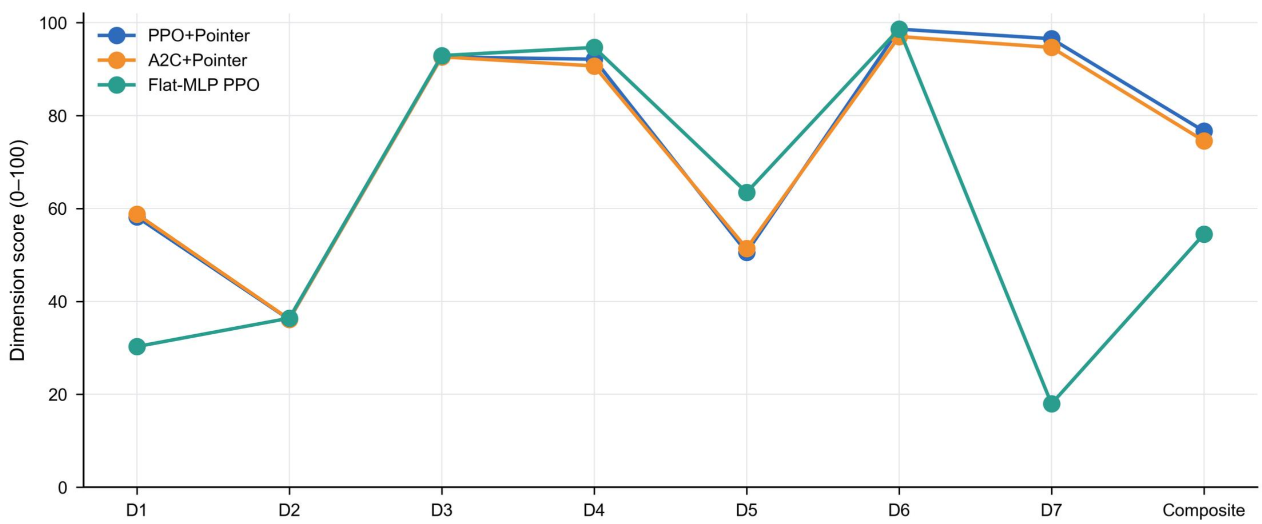

Fig. S6. Seven-dimensional operational evidence profile. The 100-point aggregate is post-hoc and not a confirmatory champion metric.

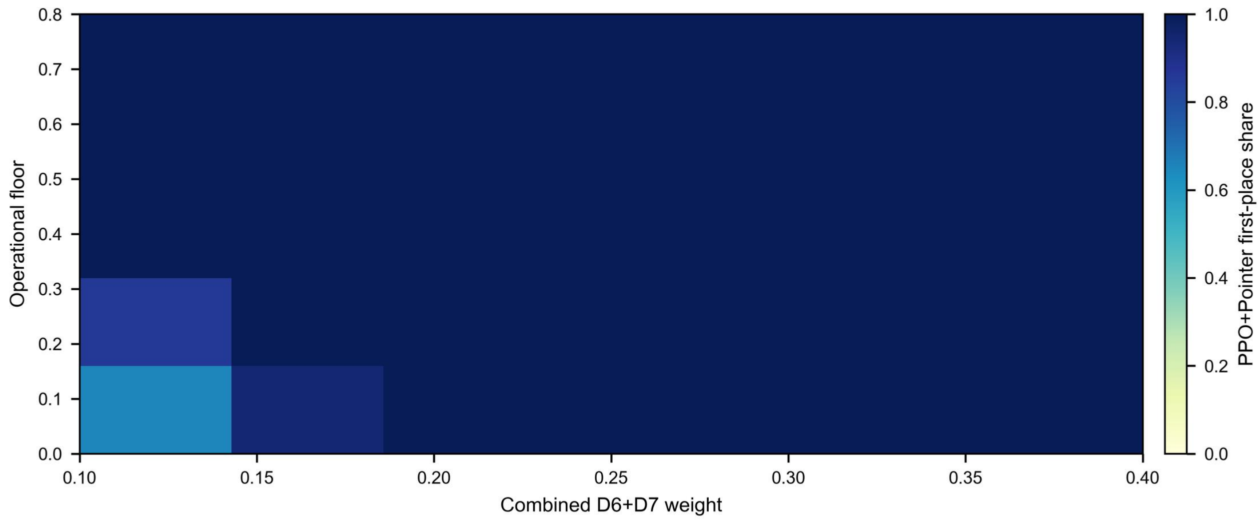

Fig. S7. Sensitivity of PPO+Pointer ranking to the operational floor and combined D6+D7 training weight.

S6. Representative unseen synthetic route

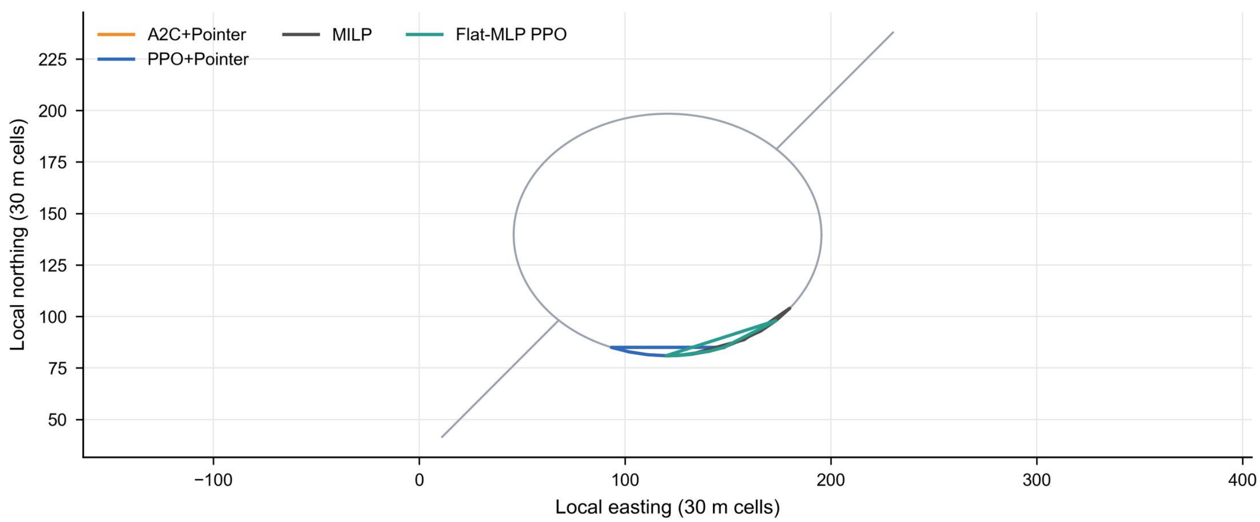

Fig. S8. Representative fixed unseen synthetic task with road context, inspection priorities, and comparator routes.

S7. Additional claim boundaries

- The geographic evidence is DSM-based simulation, not flight validation.
- All node counts were observed during training; scale extrapolation is untested.
- The ablations do not isolate energy, distance, time, and dynamics submasks separately.
- The aircraft and wind models are engineering proxies, not certified digital twins.
- Classical solver status and runtime must be interpreted jointly with route quality.
- The 100-point score is a post-hoc summary and is not used in the abstract or primary conclusion.